

Class #6 - Discrete Dynamical Systems

1. Example: In a certain region, about 4% of the city's population moves to the surrounding suburbs each year and about 3% of the suburban population moves into the city. In 2026, there were 600,000 residents in the city and 400,000 in the suburbs. Set up a difference equation that describes this situation, where x_0 is the initial population in 2026. Then estimate the population in the city and in the suburb two years later, in 2028. (Ignore other factors that might influence the population size.)

2. Example: A Predator-Prey System

In the redwood forests of California, wood rats provide up to 80% of the diet for the spotted owl, the main predator of the wood rat. Though unrealistic, a linear model can be set up as follows:

k : time in months

O_k : the number of owls

R_k : the number of rats (measured in thousands)

- (1) With no rats for food, only half of the owls will survive each month.
- (2) With no owls as predators, the rat population will grow by 10% each month.
- (3) If rats are plentiful, the owl population will rise by $0.4R_k$.
- (4) Each month, $0.104O_k$ many of rats are eaten by owls.

If we get two distinct real eigenvalues λ_1 and λ_2 with $|\lambda_1| > 1$ and $|\lambda_2| < 1$, and eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ respectively, then

For example, if the eigenvalues are 1.05 (with the corresponding eigenvector $(6, 13)$) and 0.55: Both populations will grow at _____per year. Eventually, there will be approximately _____spotted owls to every _____(thousand) rats.

For example, if the eigenvalues are 0.9 and 0.7:

3. Example: Suppose the eigenvalues of a 3×3 matrix A are 3, $4/5$, and $3/5$, with corresponding eigenvectors $\begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 1 \\ -5 \end{bmatrix}$, and $\begin{bmatrix} -2 \\ -5 \\ 3 \end{bmatrix}$. Let $x_0 = \begin{bmatrix} -2 \\ -5 \\ 3 \end{bmatrix}$. Find the solution of the equation $x_{k+1} = Ax_k$ for the specified x_0 , and describe what happens as $k \rightarrow \infty$.